



Dark Matter and Dark Energy

Common Misconceptions

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University of Cambridge

2nd October 2013

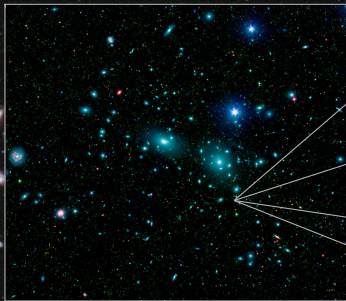
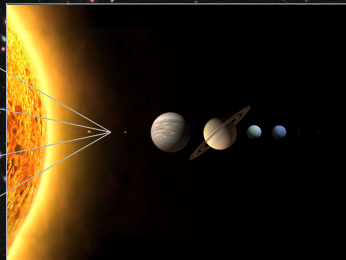
Overview

Our Picture of the Universe

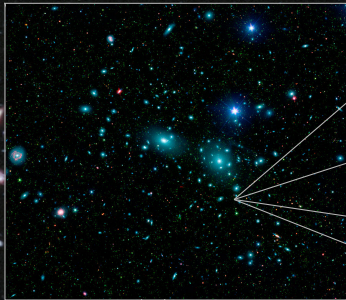
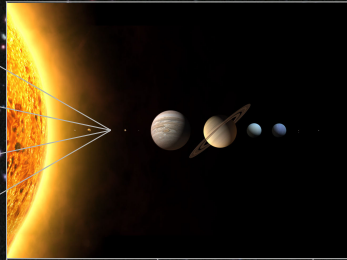
Dark Matter

Dark Energy

Our Picture of the Universe



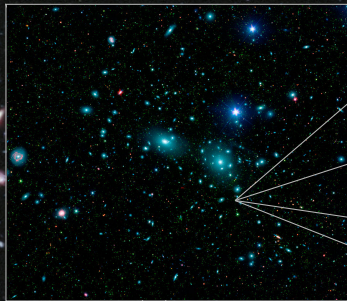
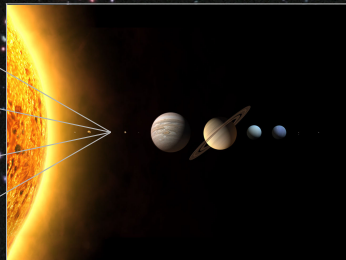
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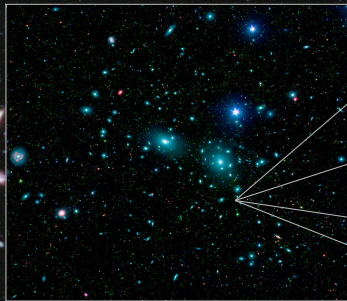
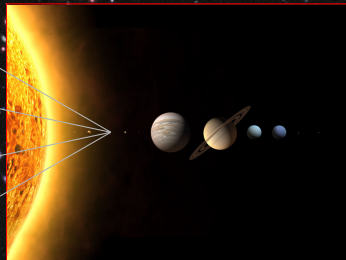
Earth



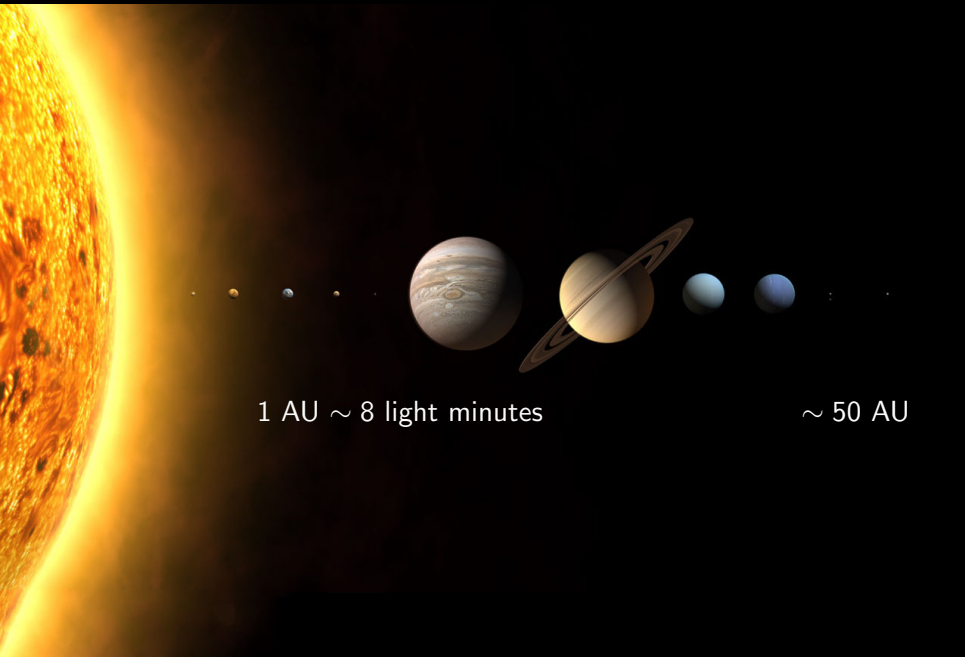
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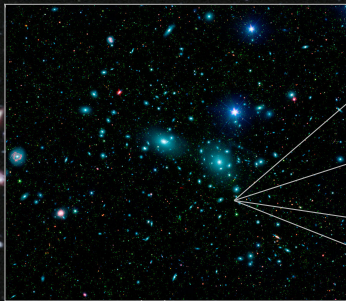
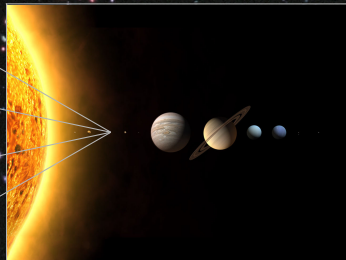
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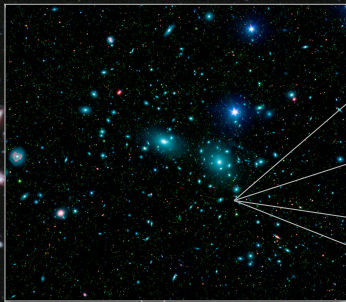
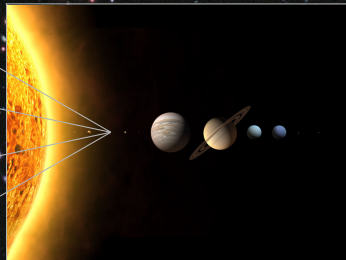
The Solar System



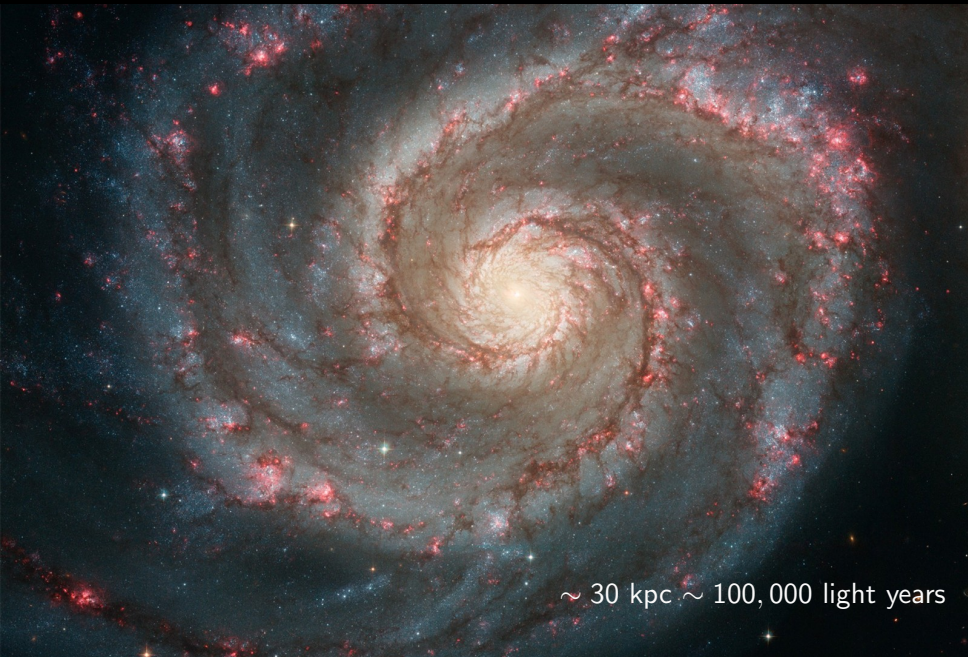
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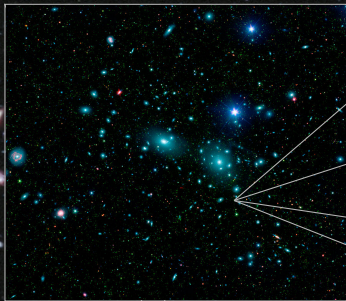
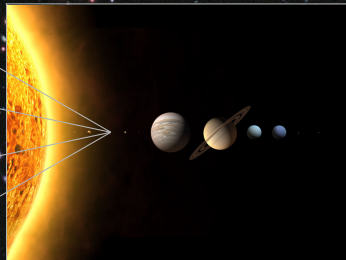


Galaxies

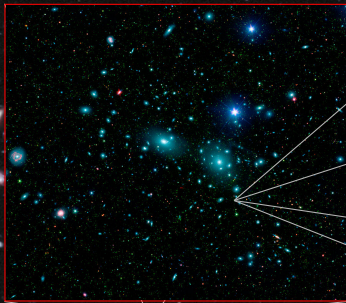
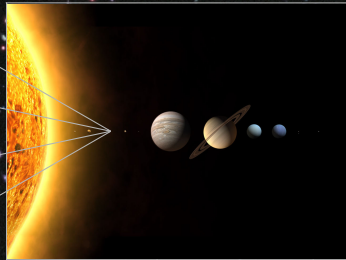


~ 30 kpc ~ 100,000 light years

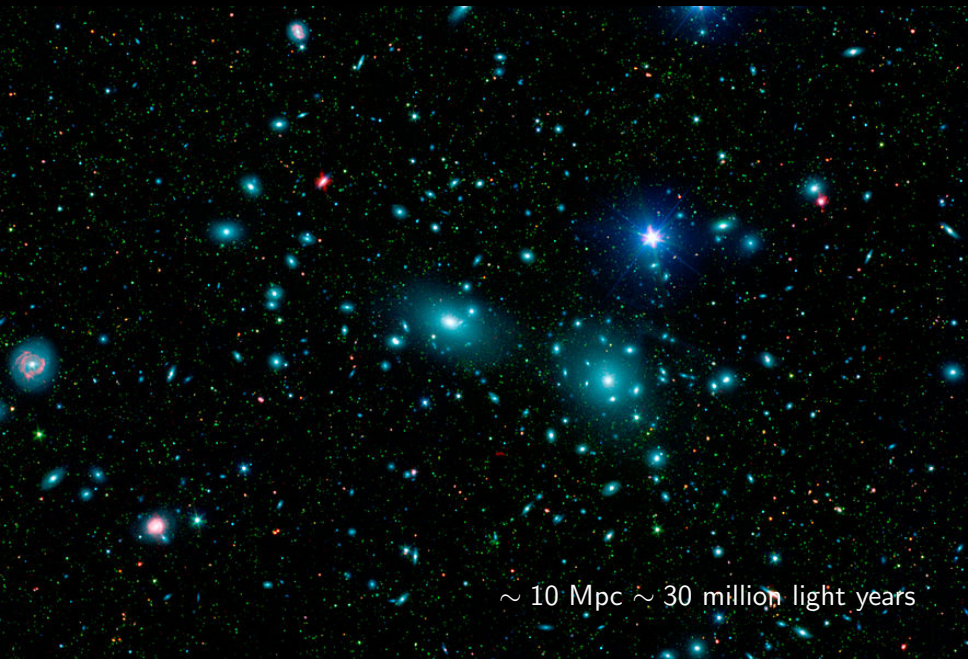
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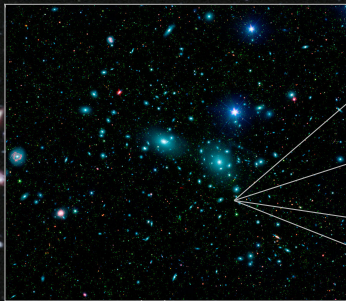
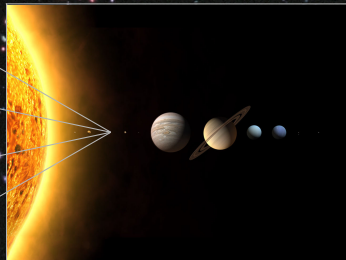


Galaxy Clusters

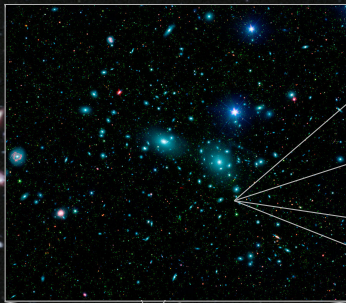
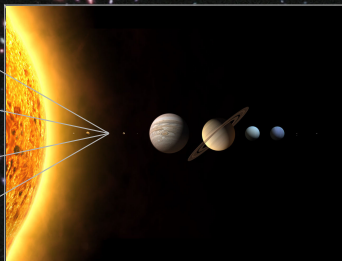


~ 10 Mpc ~ 30 million light years

Our Picture of the Universe



Our Picture of the Universe



Galaxy Clusters

Hubble extreme deep field



Dark Matter

What's missing?



Dark Matter

What's missing?

- ▶ Galaxy cluster: gravitationally bound.



Dark Matter

What's missing?

- ▶ Galaxy cluster: gravitationally bound.
- ▶ Count the matter we can see.



Dark Matter

What's missing?

- ▶ Galaxy cluster: gravitationally bound.
- ▶ Count the matter we can see.
- ▶ There is not enough matter.



What's missing

The Problem



What's missing

The Problem

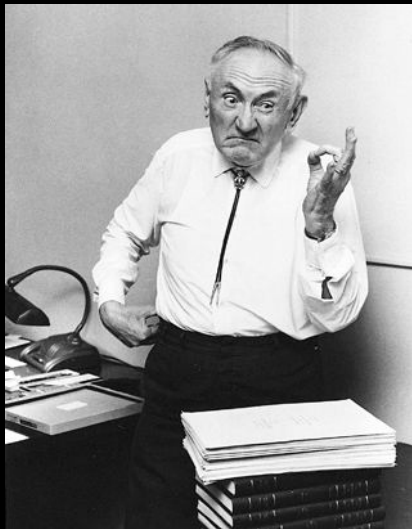


► Not enough stuff here either

Dark Matter

The Problem

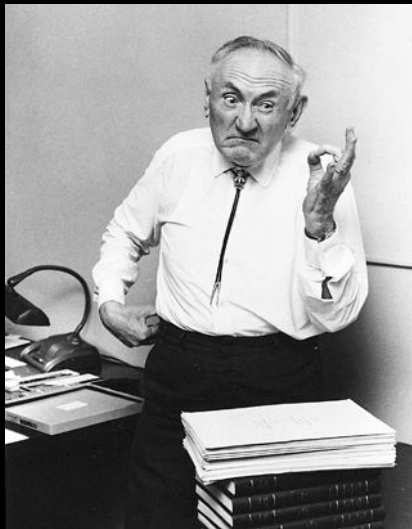
- Observed by Fritz Zwicky in 1933.



Dark Matter

The Problem

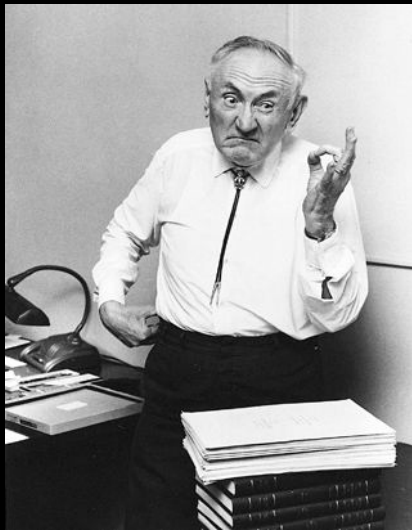
- ▶ Observed by Fritz Zwicky in 1933.
- ▶ There appears to be a large amount of matter which we cannot see.



Dark Matter

The Problem

- ▶ Observed by Fritz Zwicky in 1933.
- ▶ There appears to be a large amount of matter which we cannot see.
- ▶ We inferred its existence from its *gravitational presence*



Dark Matter as a prediction

In general, if you encounter a discrepancy between theory and observation, there are two options:

Dark Matter as a prediction

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- ▶ Modify the theory

Dark Matter as a prediction

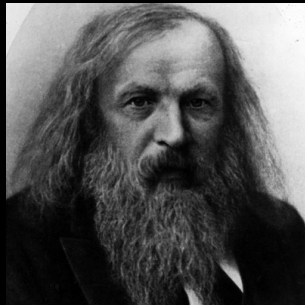
In general, if you encounter a discrepancy between theory and observation, there are two options:

- ▶ Modify the theory
- ▶ Predict a new object

Example 1.

The Periodic Table

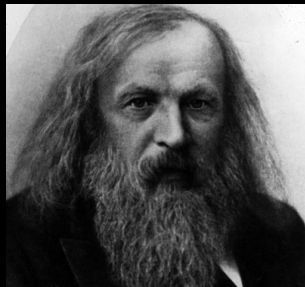
- ▶ In 1865, Mendeleev proposed that elements be ordered into the periodic table.



Example 1.

The Periodic Table

- ▶ In 1865, Mendeleev proposed that elements be ordered into the periodic table.
- ▶ He left gaps in the table, where his theory predicted there should be new matter.

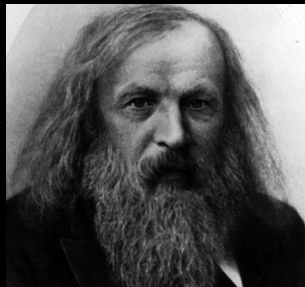


	Gruppe I. — H ⁰	Gruppe II. — H ⁰	Gruppe III. — H ⁰	Gruppe IV. H ¹ H ⁰	Gruppe V. H ¹ H ⁰	Gruppe VI. H ¹ H ⁰	Gruppe VII. H ¹ H ⁰	Gruppe VIII. — H ⁰
1	H=1							
2	Li=7	Be=9,4	B=11	C=12	N=14	O=16	F=19	
3	Na=23	Mg=24	Al=27,5	Si=28	P=31	S=32	Cl=35,5	
4	K=39	Ca=40	—=44	Ti=48	V=51	Cr=52	Mn=55	Fe=56, Co=59, Ni=59, Cu=63.
5	(Ca=63)	Zn=65	—=68	—=72	As=75	Se=78	Br=80	
6	Rb=85	Str=87	Yt=88	Zr=90	Nb=94	Mo=96	—=100	Ra=104, Rn=104, Po=106, Ag=108.
7	(Ag=108)	Cd=112	In=115	Su=118	Sh=122	Tu=125	J=127	
8	Cs=133	Ba=137	Pd=138	Ce=140	—	—	—	
9	(—)	—	—	—	—	—	—	
10	—	—	—	—	—	—	—	
11	(Au=197)	Hg=200	Tl=204	Pb=207	Bi=208	—	—	
12	—	—	—	Th=231	—	U=240	—	

Example 1.

The Periodic Table

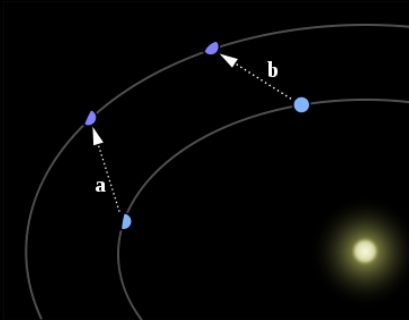
- ▶ In 1865, Mendeleev proposed that elements be ordered into the periodic table.
- ▶ He left gaps in the table, where his theory predicted there should be new matter.
- ▶ When these elements were discovered, he was hailed as a scientific hero.



Atom Nr.	Gruppe I. — H ⁰	Gruppe II. — H ⁰	Gruppe III. — R ⁰	Gruppe IV. — R ⁰	Gruppe V. — R ⁰	Gruppe VI. — R ⁰	Gruppe VII. — R ⁰	Gruppe VIII. — R ⁰
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9	(—)	—	—	—	—	—	—	
10	—	—	YEr=178	La=180	Ta=182	W=184	—	Os=195, Ir=197, Pt=198, Au=199.
11	(Au=199)	Hg=200	Tl=204	Pb=207	Bi=208	—	—	
12	—	—	—	Th=231	—	U=240	—	

Example 2.

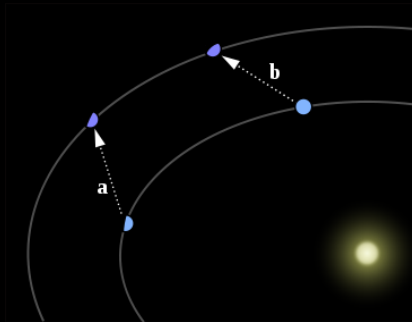
Orbits of Uranus and Mercury



- In 1841, it was noticed that Uranus' orbit was slightly off from predictions.

Example 2.

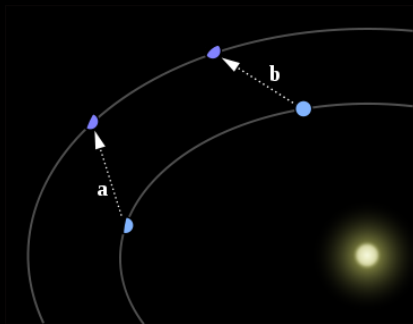
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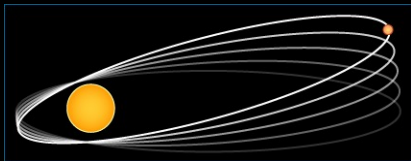
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- ▶ This observation led to the discovery of Neptune in 1846.

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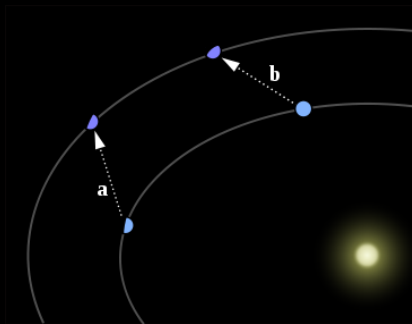
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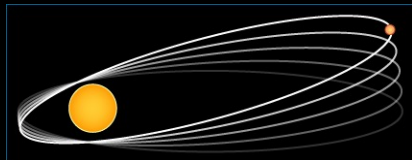
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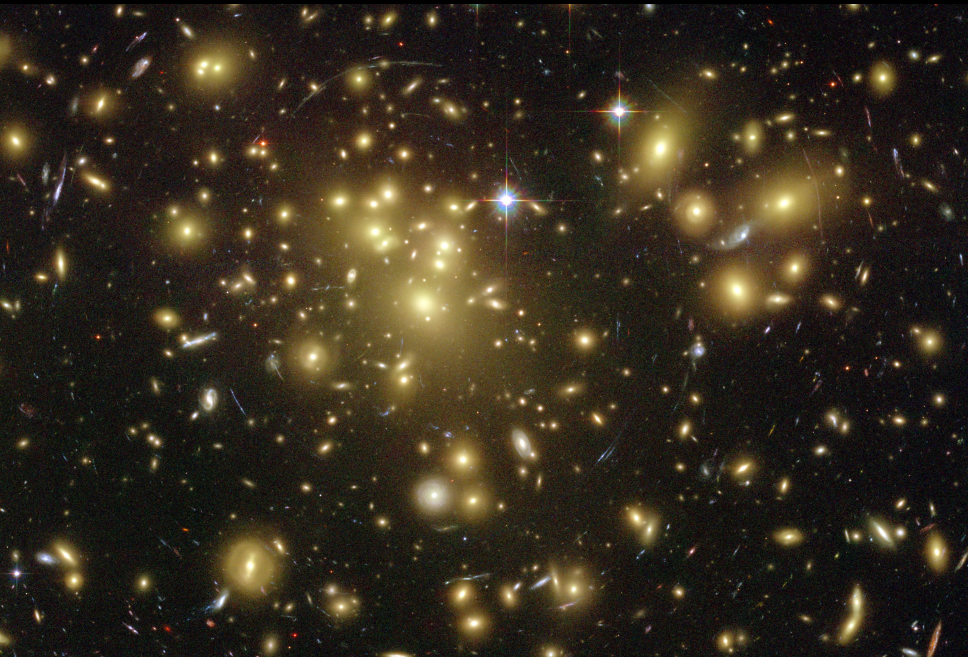


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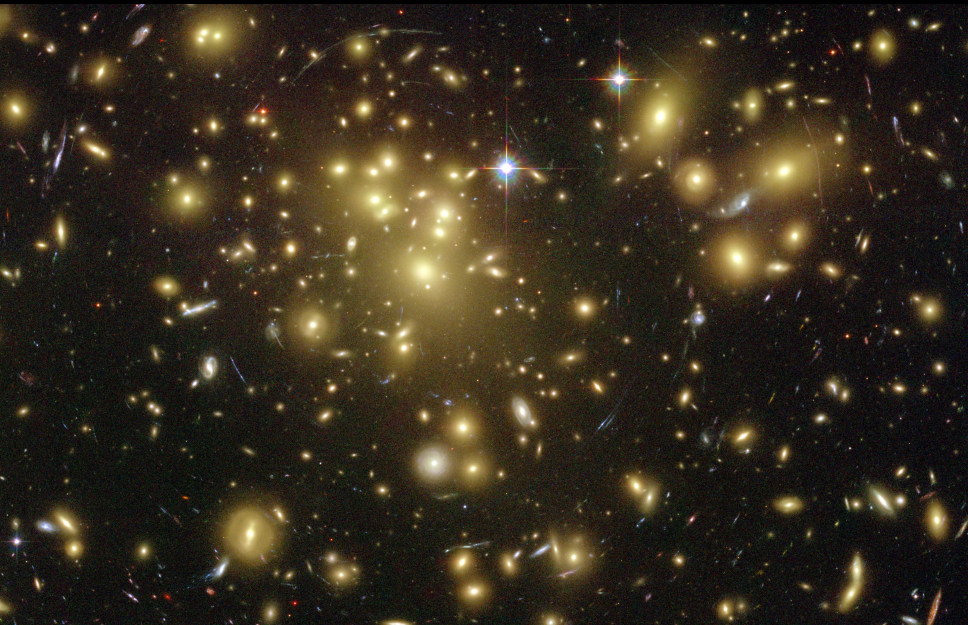
- ▶ In 1859, a similar problem emerged with the orbit of mercury
- ▶ This issue wasn't solved until Einstein's theory of general relativity superseded Newton's gravity in 1915.

Other evidence

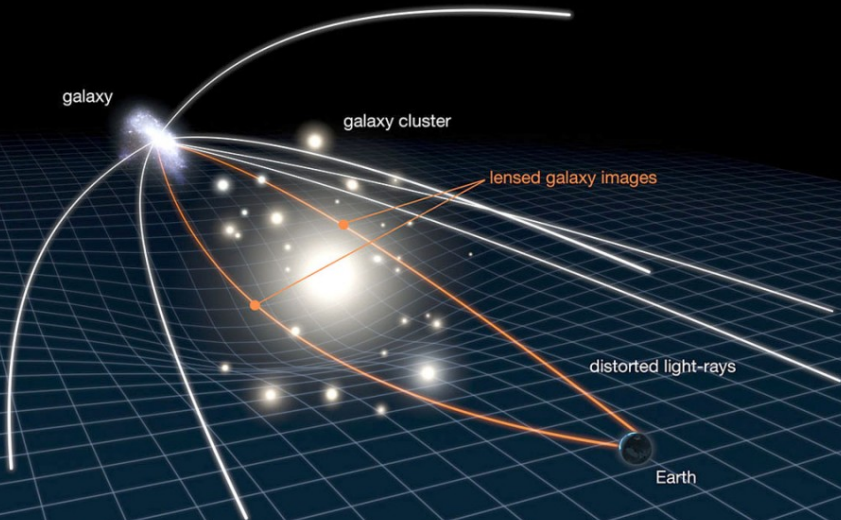


Other evidence

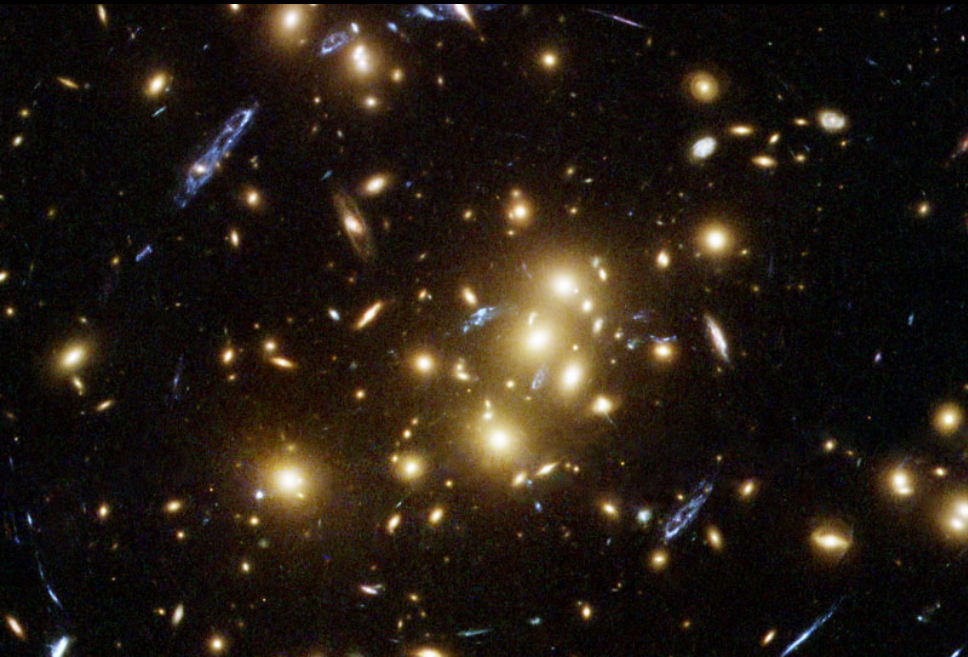
Gravitational Lensing



Gravitational Lensing



Gravitational Lensing

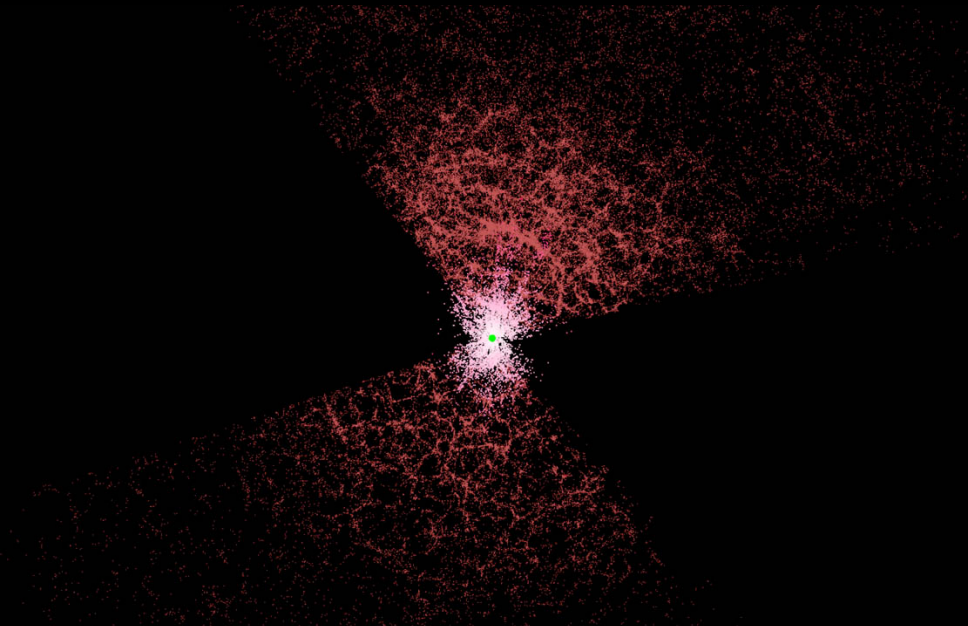


Other evidence

Structure formation

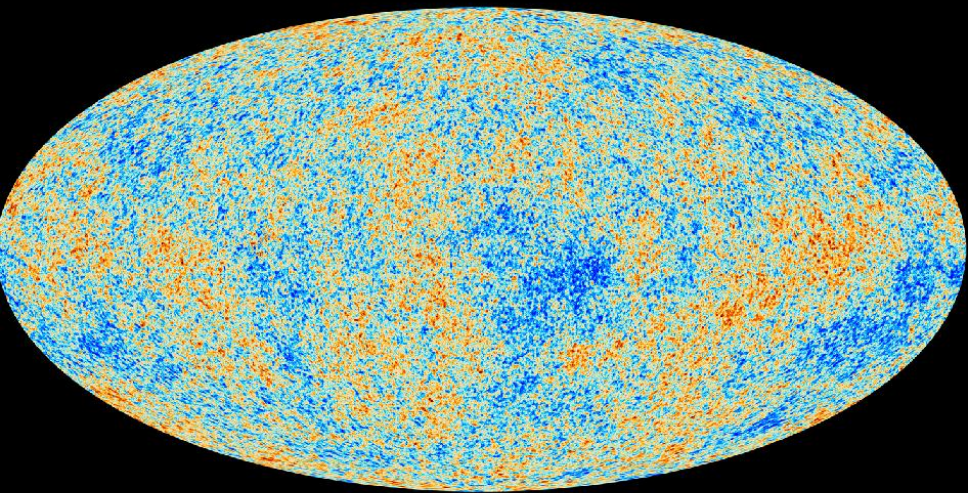
Other evidence

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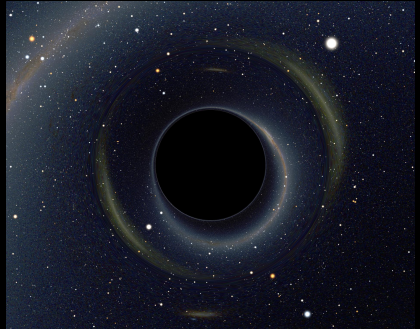
Other evidence

Cosmic Microwave Background



What could dark matter be?

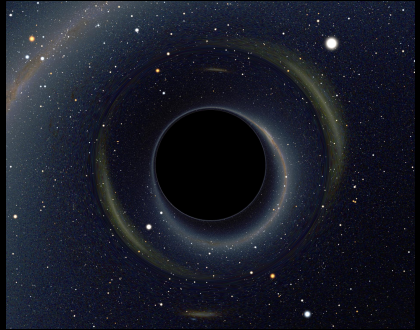
MACHOs: Massive
Astrophysical
Compact Halo
Objects.



What could dark matter be?

MACHOs: Massive
Astrophysical
Compact Halo
Objects.

WIMPs: Weakly Interacting
Massive Particles.



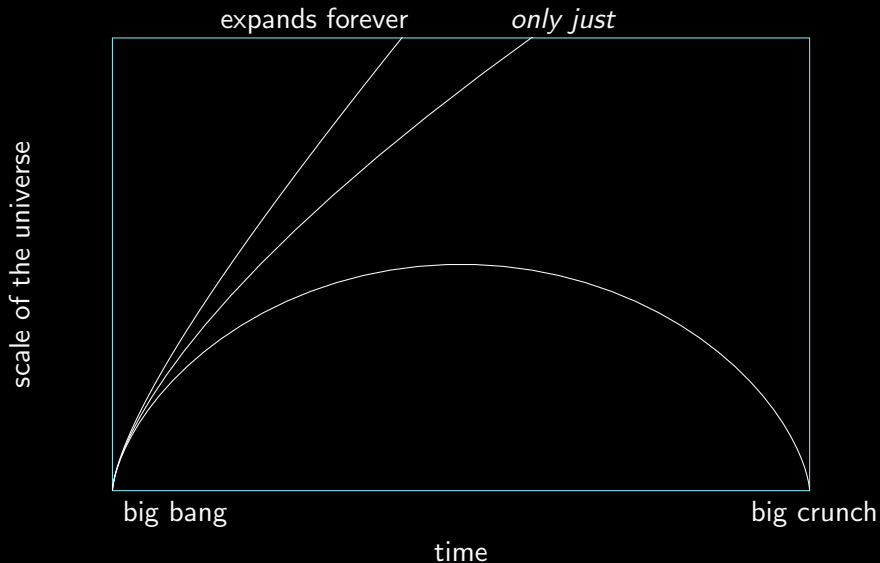
Dark Energy

The Problem



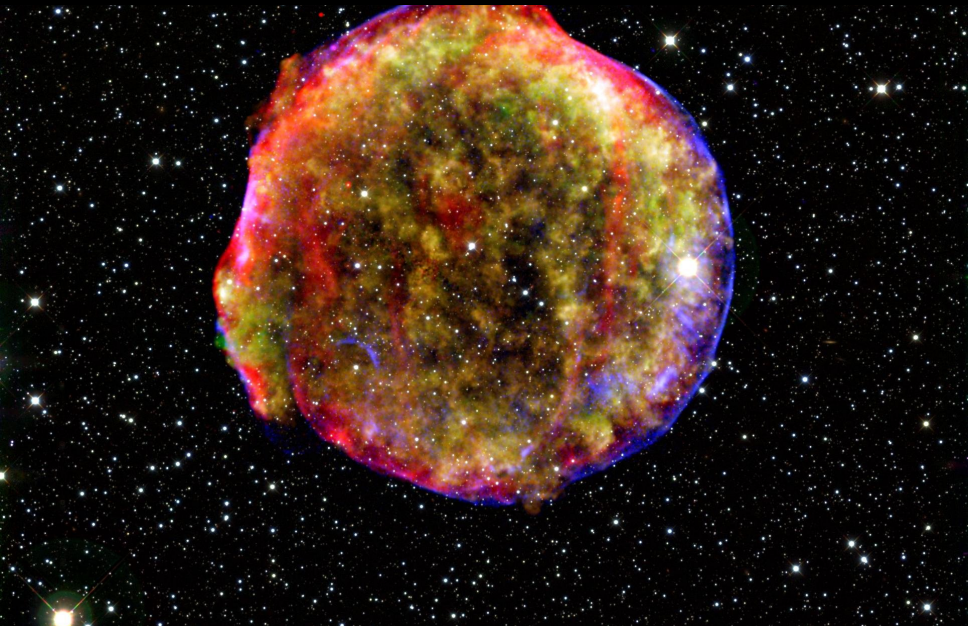
The evolution of the universe

Three options:



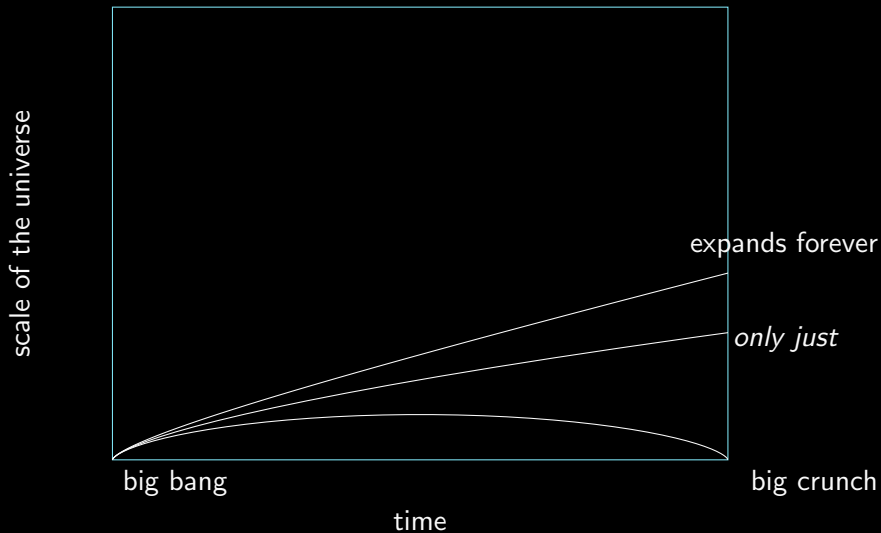
Dark Energy

The Problem



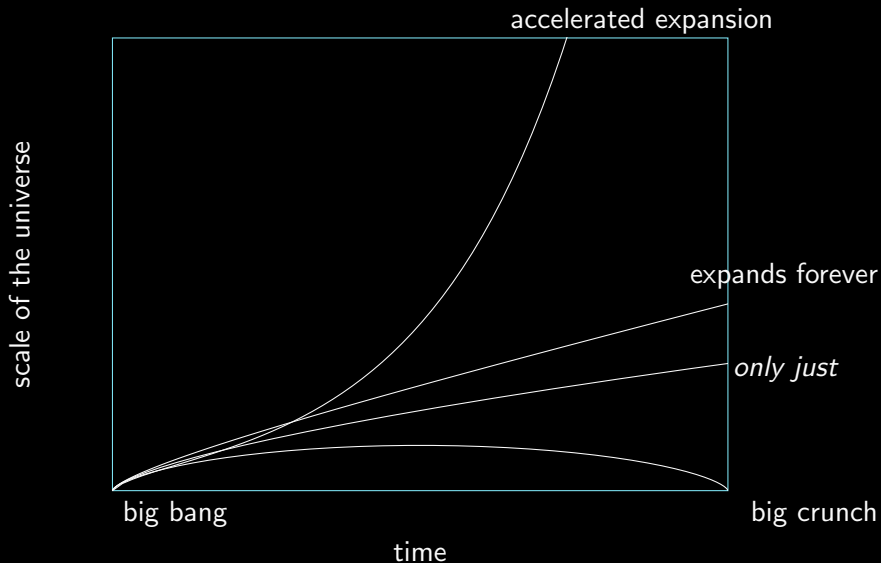
The evolution of the universe

Three options:



The evolution of the universe

None of the above:



Dark Energy

The Problem

- ▶ The Universe isn't slowing down in its expansion, but *speeding up!*

Dark Energy

The Problem

- ▶ The Universe isn't slowing down in its expansion, but *speeding up!*
- ▶ One of the ways in which this could happen would be the presence of an energy permeating the universe, accelerating the expansion.

Dark Energy

The Problem

- ▶ The Universe isn't slowing down in its expansion, but *speeding up!*
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- ▶ We call this Dark Energy

Dark Energy

The Problem

- ▶ The Universe isn't slowing down in its expansion, but *speeding up!*
- ▶ One of the ways in which this could happen would be the presence of an energy permeating the universe, accelerating the expansion.
- ▶ We call this Dark Energy
- ▶ Like Dark Matter, it's really just a name for a phenomenon which we observe, and haven't been able to explain yet

Dark Energy...

...or Cosmological Constant?

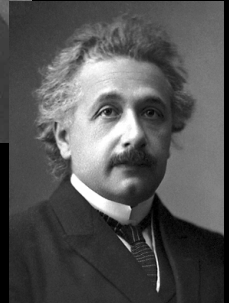
- ▶ Einstein's gravity can be repulsive at large scales.



Dark Energy...

...or Cosmological Constant?

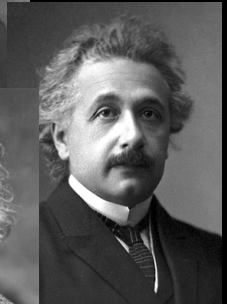
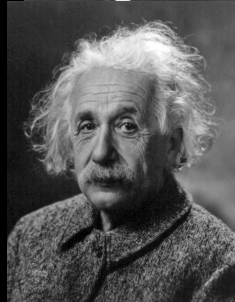
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Dark Energy...

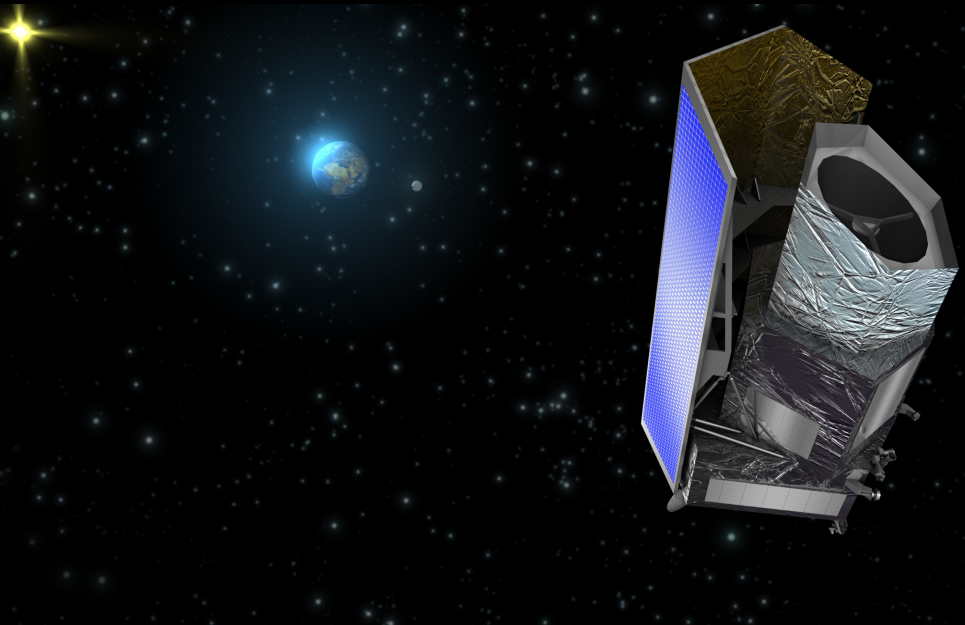
...or Cosmological Constant?

- ▶ Einstein's gravity can be repulsive at large scales.
- ▶ This is known as a cosmological constant.
- ▶ In modern terms, we think of it as a type of energy.



The Euclid Satellite

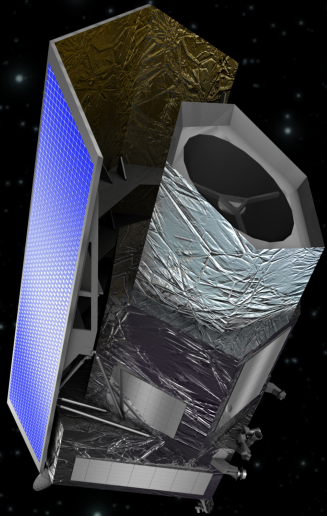
Exploring dark energy



The Euclid Satellite

Exploring dark energy

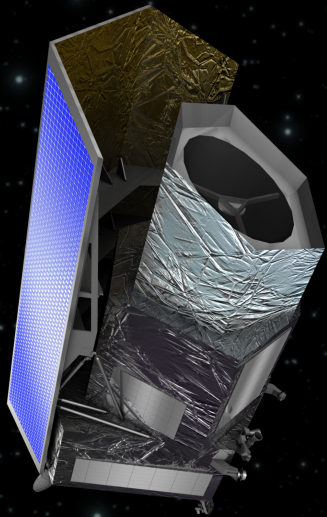
► Launch date $\approx$ 2020



The Euclid Satellite

Exploring dark energy

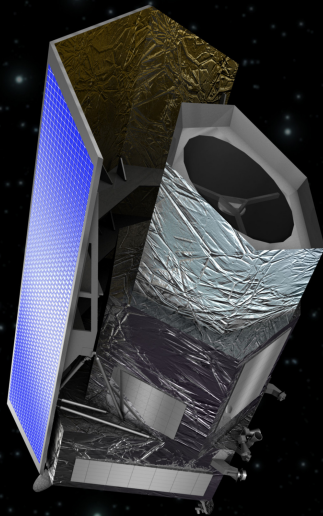
- ▶ Launch date $\approx$ 2020
- ▶ Build up an accurate 4D picture of the visible universe



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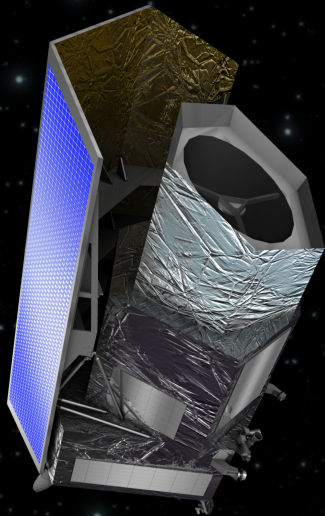
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- ▶ By measuring cosmic shear around galaxies and clusters, will be able to build up a map of Dark Matter



The Euclid Satellite

Exploring dark energy

- ▶ Launch date $\approx$ 2020
- ▶ Build up an accurate 4D picture of the visible universe
- ▶ By measuring cosmic shear around galaxies and clusters, will be able to build up a map of Dark Matter
- ▶ Will see any variation in dark energy across space and time, enabling us to build a better understanding of it.



The Composition of the Universe

